



Modelling Financial Markets

Seminar 0: Financial Markets in 2026 — Hands-on



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Acronyms used in this material

- AI: Artificial Intelligence
- API: Application Programming Interface
- AR: AutoRegressive (model)
- BET: Bucharest Exchange Trading (index)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAGR: Compound Annual Growth Rate
- CAR: Cumulative Abnormal Return
- DCC: Dynamic Conditional Correlation
- ETF: Exchange-Traded Fund
- FRED: Federal Reserve Economic Data
- FX: Foreign eXchange
- MAD: Median Absolute Deviation
- SE: Standard Error
- SR: Sharpe Ratio
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)

Outline

Objectives and Setup

Definitions Used Today

Getting and Cleaning Data

Part A: Pen-and-Paper Exercises

Part B: Computational Exercises

Part C: Open Question

Team Project

Seminar Objectives

By the end of this seminar you will be able to

- ▣ load daily market data and public online data through one function, and clean them
- ▣ derive when annualisation rules and correlation estimates fail
- ▣ attach standard errors and confidence intervals to returns, Sharpe ratios and correlations
- ▣ test whether a change is real: Fisher z, bootstrap, a first event study
- ▣ set up the GitHub repository of your team project

- ▣ Notebook for today: [Open the seminar notebook in Colab](#)
- ▣ Seminar: Drd. Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Setup: Colab, Python and the Repository

```
# in a terminal (or a Colab cell with !)  
git clone https://github.com/danpele/MFM.git  
cd MFM  
ls data/market | head      # one CSV per symbol  
  
# Python packages used in Chapter 0  
pip install pandas numpy matplotlib scipy
```

- Colab needs no installation: open the notebook link and run all cells
- data/market: daily files with open, high, low, close, adjusted close, volume
- Quantlets/Ch_00: the code behind every chart of the lecture
- notebooks/EN, notebooks/R0: lecture and seminar notebooks

GitHub Workflow for the Team Project

One repository per team

- ▣ One member creates a public repository and invites the others
- ▣ README.md: research question, data, how to run
- ▣ One folder per result, Quantlet style: code + Metainfo.txt + chart

▣ Example of the Quantlet format: [Quantlets/Ch_00/MFM_ch0_cross_asset](#)

Daily routine

- ▣ `git pull` before you start working
- ▣ Small commits with clear messages
- ▣ `git push` at the end of each session
- ▣ Never commit passwords or API keys

How This Seminar Works

- ▣ Two kinds of problems in every part
 - ▶ **[Solved]**: complete worked solutions, in the slides and in the notebook (code and results); use them as models
 - ▶ **[Proposed]**: you solve them; each one names its solved model; solutions are discussed in class
- ▣ **Part A**: pen-and-paper (A1–A4 warm-up, A5–A7 derivations); **Part B**: verification and inference on real data (Ex1–Ex8)
 - ▶ not “compute and compare with the lecture”, but “how sure are we, and when does the method fail?”
 - ▶ every exercise ends with interpretation questions that a number alone cannot answer
- ▣ **Part C**: an open question with no single method, a possible start for your team project (C1), and an AI answer to critique (C2)

Solved and Proposed Problems

Solved (models)

- ▣ A1–A2, A3–A4: returns, drawdown, annualisation, CAGR
- ▣ A5: volatility drag (derivation)
- ▣ Ex1: cross-asset dashboard; Ex1 Extended: standard errors and bootstrap
- ▣ Ex3: correlations and calendars
- ▣ Ex6: finding bad ticks
- ▣ Ex7: event study, Banca Transilvania

Proposed (model in brackets)

- ▣ A6 (A3–A4, A5), A7 (A5, Ex3)
- ▣ Ex2 drawdowns (A2, Ex1); Ex4 stablecoins (A4, Ex1); Ex5 BVB in EUR (Ex1)
- ▣ Ex3 Extended: has the correlation changed? (Ex1 Extended)
- ▣ Ex6 Extended: a calibrated threshold (Ex6)
- ▣ Ex8: event study, BRD (Ex7)
- ▣ C1: open question (Ex3, Ex3 Extended)
- ▣ C2: critique an AI answer (A3–A4, Ex1)

Definitions Used Today (1/4): Returns, Growth, Sharpe Ratio

- ▣ Simple return $R_t = P_t/P_{t-1} - 1$; log return $r_t = \ln(P_t/P_{t-1})$, with P_t the closing price on day t
- ▣ **Adjusted close**: price corrected for dividends and splits, used for stocks and ETFs
- ▣ **CAGR** (compound annual growth rate): $(P_{\text{end}}/P_{\text{start}})^{1/Y} - 1$, $Y = \text{calendar days} / 365.25$
 - ▶ $\text{CAGR} = e^{\mu_{\log}} - 1$, with $\mu_{\log}$ the annualised mean of log returns (geometric growth)
- ▣ **Arithmetic mean** μ_{arit} : annualised mean of simple returns; **growth multiple**: $P_{\text{end}}/P_{\text{start}}$
- ▣ **Sharpe ratio** (SR): $\sqrt{q} \bar{R}/s_R$, the mean over the standard deviation of daily simple returns, with the risk-free rate $r_f = 0$
 - ▶ $q = \text{observed trading days per year}$: about 252 for equities, 365 for crypto, about 261 for spot gold (weekday quotes only)
- ▣ **Year-on-year (YoY) growth**: $X_{\text{year end}}/X_{\text{previous year end}} - 1$

Definitions Used Today (2/4): Risk

- ▣ **Daily volatility:** the standard deviation $\hat{\sigma}_d$ of daily log returns
- ▣ **Annualised volatility:** $\hat{\sigma}_d\sqrt{q}$ (the " $\sqrt{P}$ rule"), exact only for uncorrelated returns (Exercise A6)
- ▣ **Autocorrelation** $\rho_j = \text{Corr}(r_t, r_{t-j})$: how today's return is related to the return j days earlier
- ▣ **Running peak:** $M_t = \max_{s \leq t} P_s$; **drawdown:** $DD_t = P_t/M_t - 1$
- ▣ **Maximum drawdown:** $\min_t DD_t$, the worst loss from a peak
- ▣ **Drawdown spell:** starts on the peak date and ends on the first close at or above that peak (full recovery)
- ▣ **Peak-to-trough decline:** trough/peak $- 1$ for a given episode

Definitions Used Today (3/4): Co-Movement and Data Terms

- **Correlation:** $\rho = \text{Cov}(r^X, r^Y) / (\sigma_X \sigma_Y) \in [-1, 1]$, computed on returns, never on prices
- Two estimators that answer different questions
 - ▶ **correlation over the whole period:** one number for all days in the period
 - ▶ **mean of the 252-day rolling correlation:** average of one-year windows (used in the lecture chart)
- Three ways to align two calendars
 - ▶ **common days:** join the *prices* on the days both assets trade, then compute returns (Friday → Monday for both)
 - ▶ **forward-fill:** repeat the last equity price on weekends, so its weekend returns are zero
 - ▶ **weekly returns:** Friday close to Friday close for both assets
- **Bad tick:** a wrong price in a data file, usually a spike reversed the next day
- **Rolling median:** the median of a moving window, e.g. 11 days centred on day t ; robust to single outliers
- **pp:** percentage point, the unit for differences between two rates in %

Definitions Used Today (4/4): Inference Tools

- **Standard error (SE)**: the standard deviation of an estimator across samples; 95% **confidence interval** $\approx$ estimate ± 1.96 SE
- **iid**: independent and identically distributed; most textbook formulas assume it
- **Stationary bootstrap** ([Politis & Romano \(1994\)](#)): resample blocks of consecutive days of random length (mean b) to keep the dependence in the data
- **Fisher z**: $\text{atanh}(\hat{\rho})$ approximately follows the Normal distribution with variance $1/(n-3)$ for iid data ([Fisher \(1915\)](#))
- **MAD** (median absolute deviation): median of $|x - \text{median}|$; $1.4826 \times \text{MAD}$ estimates σ under normality
- **Hampel filter** ([Hampel \(1974\)](#)): flag x_t if it is more than $c \times 1.4826 \times \text{MAD}$ from the rolling median
- **Market model**: $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$
 - ▶ **abnormal return** $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$; **CAR**: cumulative abnormal return over an event window

Sources Used Today

- ▣ **FRED** (St. Louis Fed): public US macro, interest-rate and exchange-rate series
- ▣ **BNR**: publishes the official EUR/RON reference rate (fixing at about 13:00)
- ▣ **DefiLlama**: a public data platform for DeFi, including stablecoin supply
- ▣ **SPY**: S&P 500 ETF; **TLT**: iShares 20+ Year Treasury Bond ETF (long US government bonds)
- ▣ **BET**: the Bucharest Stock Exchange benchmark index; **BET-TR**: its version with reinvested dividends
- ▣ **Stablecoin supply**: the total value of USD-pegged tokens in circulation, in bn USD
- ▣ **S&P 500**: 500 large US companies; Euro Stoxx 50 and Nikkei 225: euro-area and Japanese blue-chip indices

One Function for All Data

```
URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')

def fetch_daily(symbol, source='market',
               field='close'):
    if source == 'market':
        d = pd.read_csv(URL + symbol + '.csv',
                        index_col='date',
                        parse_dates=True)
        return d[field].dropna()
    if source == 'fred':
        u = ('https://fred.stlouisfed.org/graph/'
            'fredgraph.csv?id=' + symbol)
        return pd.read_csv(u, index_col=0,
                          parse_dates=True, na_values='.')\
            .iloc[:, 0].dropna()
    raise ValueError(source)

spx = fetch_daily('GSPC.INDX')
spy = fetch_daily('SPY.US', field='adjusted_close')
```

- One entry point: the rest of the code never knows where a file comes from
- `index_col='date'`: the same call as in the lecture code
- `close` for indices, FX, crypto, yields; `adjusted_close` for stocks and ETFs
- Public sources: FRED, BNR, DefiLlama

Cleaning Checklist

Calendars

- ▣ Equities: about 252 trading days a year; crypto: 365; spot gold: about 261 (weekend prints removed)
- ▣ Joint analysis: join the prices on common dates first, then compute returns
- ▣ Or aggregate to weekly returns (Friday to Friday)

▣ Exercises 6 and 6 extended apply this checklist to the EUR/RON market file

Values

- ▣ Adjusted vs. raw prices (dividends, splits)
- ▣ Weekend rows in FX or yield files
- ▣ Bad ticks: points far from a rolling median that reverse the next day
- ▣ Missing values: never fill returns with zeros silently

Exercises A1–A2 [Solved]: Returns and Drawdown, Step by Step

Tasks

- ▣ **A1:** $P_0 = 100 \rightarrow P_1 = 110 \rightarrow P_2 = 99$; simple and log returns; compare their sums with the two-period return
- ▣ **A2:** prices 100, 120, 90, 110, 130, 100; running peak, drawdown, maximum drawdown

Solution

- ▣ **A1:** R : +10%, −10%, sum 0, but the total is −1%;
 r : 9.53%, −10.54%, sum −1.005% = $\ln 0.99$
- ▣ **A2:** peak 100, 120, 120, 120, 130, 130; drawdown 0, 0, −25%, −8.33%, 0, −23.08%; maximum −25%

Exercises A3–A4 [Solved]: Annualisation and CAGR

Tasks

- ▣ **A3:** annualise a daily volatility of 1% for an equity index and for Bitcoin
- ▣ **A4:** the S&P 500 multiplied by 3.72 between 2 Jan 2015 and 18 Sep 2026 (11.71 years); find the CAGR

Solution

- ▣ A3: $1\% \times \sqrt{252} = 15.87\%$; $1\% \times \sqrt{365} = 19.10\%$
- ▣ A4: $3.72^{1/11.71} - 1 = 11.87\%$, the lecture's 11.9% a year

Exercise A5 [Solved]: Volatility Drag

Task

- Log returns r_t are iid; the simple return is $R_t = e^{r_t} - 1$
- (a) Expand $\ln(1 + R)$ to second order and show

$$\mu_{geo} \approx \mu_{arit} - \frac{1}{2}\sigma^2$$
- (b) Bitcoin (Chapter 1): daily log mean 0.118%, daily standard deviation 3.49%; compute the annualised log mean, the annual $\sigma^2/2$, the implied simple mean and the CAGR (365 days a year)
- (c) An asset has an annual arithmetic mean of 10% and a volatility of 60%: which CAGR do you expect?

Solution

- (a) $\ln(1 + R) = R - \frac{1}{2}R^2 + O(R^3)$; since

$$\mathbb{E}[R^2] = \sigma^2 + \mu^2 \approx \sigma^2, \mathbb{E}[\ln(1 + R)] \approx \mu - \frac{1}{2}\sigma^2$$
- (b) $0.118\% \times 365 = 43.1\%$;
 $\sigma^2/2 = 0.0349^2 \times 365/2 = 0.222$
 ▶ simple mean $43.1\% + 22.2\% = 65.3\%$ (direct: 65.2%);
 CAGR = $e^{0.431} - 1 = 53.9\%$
- (c) $0.10 - 0.60^2/2 = -0.08$, so CAGR
 $= e^{-0.08} - 1 \approx -7.7\%$
- A positive arithmetic mean does not guarantee growth of wealth: high volatility can turn it into a loss

Exercise A6 [Proposed]: When the $\sqrt{P}$ Rule Fails

Task

- ▣ Model: A3–A4 and A5 [Solved]
- ▣ r_t stationary, with variance σ^2 and autocorrelations ρ_j
- ▣ (a) Show that $\text{Var}(\sum_{t=1}^k r_t) = k\sigma^2[1 + 2\sum_{j=1}^{k-1}(1 - \frac{j}{k})\rho_j]$
- ▣ (b) AR(1), $\rho_j = \rho^j$: the limit of the ratio to $k\sigma^2$ as $k \rightarrow \infty$
- ▣ (c) EUR/RON, BNR rate (Chapter 1): $\hat{\rho}_1 = 0.17$, volatility with $\sqrt{252} = 4.4\%$; correct it assuming AR(1)
- ▣ (d) The market file with bad ticks has $\hat{\rho}_1 \approx -0.5$: in which direction is $\sqrt{252}$ wrong now?
- ▣ (e) Check (c) with monthly returns: does the AR(1) correction hold?

Solution

- ▣ *Solution discussed in class*

Exercise A7 [Proposed]: Why Is the Weekly Correlation Lower than the Daily One?

Task

- ▣ **Model:** A5 and Ex3 [Solved]
- ▣ x_t (S&P 500), y_t (Bitcoin): daily returns on common days, without own autocorrelation; cross-covariances $\gamma(j) = \text{Cov}(x_t, y_{t-j})$
- ▣ (a) For 5-day sums X, Y show $\text{Cov}(X, Y) = 5\gamma(0) + \sum_{j=1}^4 (5-j)[\gamma(j) + \gamma(-j)]$, and that the weekly correlation equals the daily one if $\gamma(j) = 0$ for $j \neq 0$
- ▣ (b) Ex3: 0.42 daily, 0.30 weekly (2022–2026); which sign must the lagged covariances have? The [Epps \(1979\)](#) effect predicts the opposite
- ▣ (c) With about 246 weeks, $\text{SE} \approx (1 - \hat{\rho}^2)/\sqrt{n}$: is the difference clearly significant?
- ▣ (d) Recompute the weekly correlation anchored on Monday, . . . , Thursday

Solution

- ▣ *Solution discussed in class*

Ex1 [Solved]: A Cross-Asset Dashboard

Task

- From data/market: S&P 500 with dividends (SPY), Euro Stoxx 50, Nikkei 225, gold, TLT, Bitcoin
- Euro Stoxx 50 and Nikkei 225 to USD with the FRED EUR/USD and USD/JPY rates
- Growth of 1 USD since 2 Jan 2015, log scale
- CAGR and annualised volatility, with q = observed trading days per year

Interpretation

- Gold: weekend prints are removed (weekday quotes only, about 261 a year); how much does annualising with 252 instead of the actual frequency change its volatility? Check with the GLD ETF

Results

- S&P 500 with dividends: $\times 4.5$, 13.7% a year, volatility 17.6%
- Nikkei 225 (USD): $\times 2.9$, 9.5%; Euro Stoxx 50 (USD): $\times 1.9$, 5.7%
- Gold, weekday quotes only: $\times 3.7$, 11.8%, volatility 16.3% with its 261 quotes a year (16.1% with 252)
- TLT: $\times 0.9$, -1.0% a year; Bitcoin: $\times 257$, 60.6%, volatility 66.7%
- GLD ETF cross-check: 16.3%; once weekend prints are removed, the choice of annualisation factor moves gold's volatility by only 0.2 pp

Ex1 Extended [Solved]: How Reliable Are the Numbers in the Table?

Task

- For each asset of Ex1, the standard error of the annual Sharpe ratio
 - ▶ iid, Lo (2002), with Y years: $SE \approx \sqrt{(1 + SR^2/2)/Y}$
 - ▶ corrected for skewness γ_3 and kurtosis γ_4 (Opdyke (2007)), with the daily SR: $Var \approx [1 + \frac{1}{2}SR^2 - \gamma_3SR + \frac{\gamma_4 - 3}{4}SR^2]/T$
- Stationary bootstrap, mean block 20 days, 2 000 replications: 95% intervals for CAGR and Sharpe
- Sensitivity: blocks of 5 and 60 days
- 95% interval for the Sharpe difference S&P 500 minus gold, on common days

Worked example

- S&P 500 with dividends: $SR = 0.82$ (mean over standard deviation), $Y = 11.71$
- $SE = \sqrt{(1 + 0.82^2/2)/11.71} = 0.34$, so the 95% interval is [0.16; 1.48]

Interpretation

- Which differences between assets survive once uncertainty is added?
- Is the iid formula too optimistic for Bitcoin? Compare it with the bootstrap
- How much do the intervals depend on the block length, and what does that say about dependence in the data?

Ex1 Extended [Solved]: Results

- Sharpe (SE Lo; SE Opdyke; bootstrap 95%, block 20)
 - ▶ S&P 500: 0.82 (0.34; 0.30; [0.32, 1.40])
 - ▶ Gold: 0.77 (0.33; 0.30; [0.23, 1.29])
 - ▶ Nikkei: 0.54 (0.31; 0.29; [0.08, 1.02])
 - ▶ Euro Stoxx: 0.37 (0.30; 0.29; [-0.16, 0.92])
 - ▶ TLT: 0.00 (0.29; 0.29; [-0.52, 0.51])
 - ▶ Bitcoin: 1.05 (0.36; 0.29; [0.44, 1.66])
- S&P 500 minus gold: +0.06, 95% interval [-0.65, +0.81]
- All intervals overlap: in 11.7 years no Sharpe ranking is established, and S&P 500 vs gold is indistinguishable
- CAGR is even less precise: S&P 500 [4.4%, 23.7%], Bitcoin [7%, 141%]
- Block length matters little: S&P 500 [0.29, 1.39] with 5 days, [0.37, 1.35] with 60
- Bitcoin: the annual iid formula (0.36) is not narrower than the bootstrap here; the daily SR is so small that fat tails barely move Opdyke's SE

Ex2 [Proposed]: Drawdowns and Recovery Time

Task

- ▣ **Model:** A2 and Ex1 [Solved]
- ▣ Drawdown series of the S&P 500 (2000–2026) and Bitcoin (Sep 2014–2026)
- ▣ Maximum drawdown and its date
- ▣ Longest spell below a previous peak; convention: the spell starts on the peak date and ends on the first close at or above the peak

Interpretation

- ▣ Bitcoin's peak date depends on the data source's daily close (00:00 UTC snapshot or another time zone): which results are sensitive to it?

Results

- ▣ *Solution discussed in class*

Ex3 [Solved]: Correlations and Calendars

Task

- Bitcoin vs S&P 500, 2022–2026, correlation over the whole period
 - ▶ daily, common days: prices joined first, then returns
 - ▶ daily, equities forward-filled on weekends
 - ▶ weekly, Friday to Friday
- Compare with the mean of the 252-day rolling correlation used in the lecture
- Stock–bond (S&P 500 vs TLT), 2010–2020 and 2022–2026, with both estimators

Results

- Bitcoin–S&P 500, whole period: common days 0.42; forward-filled 0.39; weekly 0.30
- Mean of the 252-day rolling correlation: 0.38 (the lecture's value)
- Stock–bond, whole period: -0.47 and $+0.11$; mean rolling: -0.43 and $+0.08$

Interpretation

- Why do the two estimators differ, and which one answers "what was the correlation in 2022–2026"?

Ex3 Extended [Proposed]: Has the Stock–Bond Correlation Really Changed?

Task

- ▣ Model: Ex1 Extended [Solved]
- ▣ S&P 500 – TLT: -0.47 (2010–2020) and $+0.11$ (2022–2026)
- ▣ (a) Fisher z test, valid for iid data:
$$z = \frac{\operatorname{atanh} \hat{\rho}_2 - \operatorname{atanh} \hat{\rho}_1}{\sqrt{1/(n_1 - 3) + 1/(n_2 - 3)}}$$
- ▣ (b) Stationary bootstrap in each period separately (mean block 20 days): 95% interval for $\hat{\rho}_2 - \hat{\rho}_1$
- ▣ (c) Compare the widths of the iid and bootstrap intervals
- ▣ (d) Bitcoin – S&P 500 (2022–2026): intervals for the daily and the weekly correlation

Interpretation

- ▣ Why is the bootstrap interval wider?
- ▣ Does the stock–bond change remain significant? And the daily–weekly gap of Exercise A7?
- ▣ Which economic event should explain a change of sign of the stock–bond correlation?

Results

- ▣ *Solution discussed in class*

Ex4 [Proposed]: Stablecoin Growth

Task

- ▣ **Model:** A4 and Ex1 [Solved]
- ▣ Read the total stablecoin supply online from [DefiLlama](#)
- ▣ Year-end values and year-on-year growth
- ▣ Peak-to-trough decline in 2022–2023, with exact dates
- ▣ Compound annual growth since 1 Jan 2020

Interpretation

- ▣ Supply is a stock, not a price: what does a fall in supply mean, and would the same fall in a stablecoin's price mean the same?

Results

- ▣ *Solution discussed in class*

Ex5 [Proposed]: BVB in RON and in EUR

Task

- ▣ **Model:** Ex1 (currency conversion) [Solved]
- ▣ BET-TR from data/market; official EUR/RON reference rate of the BNR
- ▣ Convert BET-TR to euro: $P_t^{EUR} = P_t^{RON} / S_t$
- ▣ CAGR and volatility in RON and in EUR since 5 Jan 2015
- ▣ Correlation of daily BET-TR returns with EUR/RON changes

Interpretation

- ▣ For a euro-based investor, is the leu's depreciation a cost or a hedge? What would change your answer?

Results

- ▣ *Solution discussed in class*

Ex6 [Solved]: Finding Bad Ticks

Task

- EUR/RON market file, 2015–2026
- Drop weekend rows; flag points more than 0.05 RON away from an 11-day centred rolling median
- Annualised volatility: raw, cleaned, BNR reference rate

Interpretation

- The cleaned series has a volatility of about 2.5%, BNR about 2.1%: why does a gap remain after the errors are removed?

Results

- 58 weekend rows and 11 weekday points removed; largest deviation 0.84 RON on 13 Aug 2025
- Volatility: 12.1% raw, 2.5% cleaned, 2.1% BNR
- The gap: the file quotes a market price at another time than the 13:00 fixing
 - ▶ daily changes correlate only 0.36 with BNR's; lag-1 autocorrelation -0.11 vs $+0.07$ (quote noise)
 - ▶ 116 days are quoted in the file but have no BNR fixing (Romanian holidays)
 - ▶ deviations smaller than 0.05 RON stay in the series

Ex6 Extended [Proposed]: A Calibrated Threshold

Task

- ▣ Model: Ex6 [Solved]
- ▣ Hampel filter: flag day t if $|x_t - m_t| > c \cdot 1.4826 \cdot \text{MAD}_t$, with m_t the median and MAD_t the MAD over a centred 11-day window
- ▣ (a) On the BNR series 2015–2026, considered clean, choose the smallest c that flags at most 0.1% of the days
- ▣ (b) Apply this c to the EUR/RON market file: how many days are flagged, and how many coincide with the 0.05 RON rule?
- ▣ (c) Volatility after cleaning, compared with 2.1% (BNR)

Interpretation

- ▣ Why MAD and not the standard deviation? Where does 1.4826 come from?
- ▣ In calm periods MAD_t is almost zero: what happens to the filter? Propose a correction

Results

- ▣ *Solution discussed in class*

Ex7 [Solved]: Event Study, the Bank Tax of December 2018

Task

- The tax was announced on the evening of 18 Dec 2018; event day $0 = 19$ Dec 2018
- Banca Transilvania, daily log returns; BET as the market
- (a) Market model $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$ on the window $[-250, -11]$
- (b) $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$ and $CAR_i[0, +5]$
- (c) $t = CAR_i / (\hat{\sigma}_{\varepsilon,i} \sqrt{L})$, with $L = 6$ days
- Method: MacKinlay (1997); Campbell, Lo & MacKinlay (1997), Ch. 4

Interpretation

- BET contains both banks with large weights: what does this do to $\hat{b}_i$ and to AR? Propose an alternative market index
- The t test assumes no change of variance in the event window: plausible here? Which test corrects it (Boehmer, Musumeci & Poulsen (1991))?
- Was there a reversal afterwards? What does it say about market efficiency (Chapter 2)?



Ex7 [Solved]: Results

- Banca Transilvania: $\hat{b} = 1.18$, $\hat{\sigma}_\varepsilon = 0.83\%$
 - ▶ day 0: $r = -22.2\%$ (log), market -11.9% , $AR_0 = -8.3\%$ ($t = -10.0$)
 - ▶ days 1–5: $+3.2\%$, $+0.6\%$, $+2.5\%$, $+0.1\%$, $+0.9\%$
 - ▶ $CAR[0, +5] = -1.1\%$, $t = -0.54$, $p = 0.59$
- Day 0 is a huge surprise, but the abnormal returns of days 1–3 reverse most of it
- The market itself fell 11.9% because of the same news: the market model removes most of the shock, so AR understates it
- Better market: BET without banks, or a regional index; variance-robust test: [Boehmer, Musumeci & Poulsen \(1991\)](#)

Ex8 [Proposed]: The Same Event for BRD

Task

- **Model:** Ex7 [Solved]
- Repeat Ex7 for BRD: same window $[-250, -11]$, BET as the market, same t test
- (a) $\hat{b}$, $\hat{\sigma}_\varepsilon$, AR_0 with its t , $CAR[0, +5]$ with t and p
- (b) Which bank was hit harder relative to its normal volatility?
- (c) Propose a variance-robust test (Boehmer, Musumeci & Poulsen (1991)) and explain why it is needed here

Interpretation

- Both banks: is the reversal a sign of overreaction or of a partly withdrawn policy? What evidence would separate the two?

Results

- *Solution discussed in class*

C1: Did Spot ETFs Change Bitcoin's Correlation with Equities?

The question

- Spot Bitcoin ETFs were approved in the US on 10 Jan 2024
- Did the Bitcoin – S&P 500 correlation rise *because of* them, or had it risen already since 2022?
- There is no single method: propose one and defend it

Deliverable

- A Quantlet folder in your team repository: code, Metainfo, one chart
- One slide with the conclusion and its main limitation

What your answer must contain

- A precise hypothesis: what would have changed, and when
- Known-date test ([Chow \(1960\)](#)) vs unknown-date test ([Andrews \(1993\)](#)): which one, and why?
- Robustness: calendar (common days, weekly), window, bootstrap block length
- A counterfactual: what happened to the correlation of other assets in the same period?
- For the curious: dynamic DCC correlation, Chapter 6

C1: Reference Analysis

- ▣ *Solution discussed in class*
- ▣ Your Quantlet is compared with a reference analysis after the deadline

C2 [Proposed]: Critique an AI Answer (1/2)

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “Using the daily files in data/market, compute the annualised mean, the annualised volatility and the Sharpe ratio ($r_f = 0$) of SPY and Bitcoin from daily simple returns”
- **The answer:** the code below, followed by the conclusion

```
import numpy as np, pandas as pd
URL = 'https://raw.githubusercontent.com/danpele/MFM/main/data/market/'
def load(sym, field='close'):
    return pd.read_csv(URL + sym + '.csv', index_col='date',
                       parse_dates=True)[field].dropna()
spy, btc = load('SPY.US'), load('BTC-USD.CC')
A = 252                                     # trading days per year
for name, p in [('SPY', spy), ('BTC', btc)]:
    r = p.pct_change().dropna()             # daily simple returns
    mu = r.mean() * A                       # annualised mean
    vol = r.std() * A                       # annualised volatility
    print(name, mu, vol, mu / vol)          # Sharpe ratio, rf = 0
```

- **Conclusion of the AI:** “Sharpe ratios: SPY 0.03, Bitcoin 0.07. Both assets barely reward their risk.”

C2 [Proposed]: Critique an AI Answer (2/2)

Task

- ▣ **Model:** A3–A4 and Ex1 [Solved]; the definitions of today
- ▣ (a) Find the errors: there are two or three; the rest of the answer is correct
- ▣ (b) For each error, say how you would detect it: a test, a published number, the lecture or the library documentation
- ▣ (c) Give the corrected code and the numbers it produces
- ▣ (d) Log each error as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- ▣ AI code usually runs without an error message: the mistakes are in the finance, not in the syntax
- ▣ Course rule: you may use AI, declare it in `AI_USE.md` and log at least three errors you caught in `AI_ERRORS.md`
- ▣ *Solution discussed in class*

Forming Your Team

Steps

- ▣ Form teams of 3–4 students
- ▣ Create the team repository on GitHub and add the instructors as collaborators
- ▣ Register the team and a tentative topic by the date announced in class
- ▣ The project is 70% of the grade: start the repository today and commit every week

A good first topic

- ▣ One market, one question, real data
- ▣ Example: do Romanian blue chips move with European banks or with energy prices?
- ▣ Example: Part C1 above, extended to Ethereum

Take-Home Assignment

Before the next seminar

- ▣ Run the seminar notebook end to end in Colab and fill in the empty cells
- ▣ Add one asset of your choice from data/market to Ex1 extended, with its Sharpe interval
- ▣ Push the result to your team repository as your first Quantlet folder
- ▣ Answer the Chapter 0 quiz on the course website
- ▣ Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Recommended courses

-  Statistics of Financial Markets
-  Cryptos, NFTs, Digital Assets
-  Introduction to Blockchain and Cryptocurrencies
-  Digital Transformation on Finance

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- ▣ Public data: [FRED](#), [BNR reference rates](#), [DefiLlama](#)
- ▣ Code and data: github.com/danpele/MFM